Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	Family Guy
Project / Solution Name	Wheat
Name (Leader)	Wu YiWei
Email & Phone (Leader)	era2509029@xmu.edu.my +60 1135449974
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://wheat-assistant.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/Family-Guy-XMUM/wheat_assistant

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Wheat is a web application for weekly wholesale order preparation, with a prototype deployed on Replit. A React/TypeScript interface communicates with a FastAPI backend. Uploaded CSV/XLSX orders, a product master and an output template are parsed by the backend. Optional AI planning sends bounded file profiles and samples to the Agnes API service using agnes-2.5-flash and returns a constrained JSON workflow. Pydantic and application checks validate the plan before employees review and approve the rules. Bundled Python operations perform the transformations without executing arbitrary model-generated code. SQLAlchemy uses PostgreSQL for cloud approvals, runs, activity and file bytes, and SQLite locally. openpyxl generates an eight-sheet workbook with explicit exceptions and row-level provenance. Replit also supported development and testing of the Status Guard module and part of the project's data analysis. Orbyte handover is simulated. Demonstrations use team-created synthetic data. No model training or fine-tuning is performed.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.12+ Local 3.12.13; cloud 3.12.12	Backend, local launcher and deterministic workflow execution.	PSF-2.0 and bundled notices
TypeScript 5.9.3	Typed frontend code and build-time type checking.	Apache-2.0
React 19.3.1 react-dom 19.3.1	Browser workspace for file review, rule approval and results.	MIT

Name & Version	Purpose in Project	License
Vite 6.4.5 @vitejs/plugin-react 4.7.0	Frontend build and React transform.	MIT
FastAPI 0.156.8	Typed HTTP API, uploads and workflow actions.	MIT
Uvicorn 0.54.0	ASGI application server.	BSD-3-Clause
Pydantic 2.14.1 pydantic-core 2.50.1	Workflow schema and input validation.	MIT
SQLAlchemy 2.1.5	SQLite / PostgreSQL persistence.	MIT
openpyxl 3.1.5	Read and export XLSX workbooks.	MIT
defusedxml 0.7.1	Safer XML parsing for workbook inputs.	PSF License
HTTPX 0.28.1	HTTP requests to the model provider.	BSD-3-Clause
python-multipart 0.0.32	Parse uploaded multipart files.	Apache-2.0
psutil 7.2.2	Process cleanup and resource measurements.	BSD-3-Clause
platformdirs 4.13.5	Local application data locations.	MIT
python-dotenv 1.2.4	Load optional local environment settings.	BSD-3-Clause
psycopg / psycopg-binary 3.3.6	PostgreSQL driver for Replit cloud storage.	LGPL-3.0-only; bundled notices
pytest 9.1.1 Hypothesis 6.170.5	Regression and generated-input tests.	MIT (pytest); MPL-2.0 (Hypothesis)
@playwright/test 1.64.1	Browser workflow acceptance.	Apache-2.0
PyInstaller 6.22.3	Build native desktop bundles.	GPL-2.0-or-later with bootloader exception
Hatchling (build requirement; unpinned)	Python package build backend.	MIT
Node.js 22+; uv (build tools)	Frontend / Python dependency installation and builds.	Node: MIT + notices; uv: MIT OR Apache-2.0
PyMuPDF 2.0	Local PDF text correction and rendering.	GNU AGPL 3.0 or commercial license
pypdf 6.10.0	Local PDF text / metadata inspection.	BSD-3-Clause
Poppler 26.04.0	Local PDF page rendering and visual QA.	GPL-2.0-only OR GPL-3.0-only

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Agnes API service OpenAI-compatible chat-completions API	Suggests column mappings and a constrained JSON workflow from bounded file profiles and samples. Proposed rules require validation and user approval before execution.	Free
N/A	N/A	N/A
N/A	N/A	N/A
N/A	N/A	N/A
N/A	N/A	N/A

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Hosts the prototype with PostgreSQL; supports Status Guard development, testing and data analysis.	Core account Reserved VM, Asia 0.5 vCPU / 2 GiB Managed PostgreSQL
GitHub	Source-code hosting and version control.	GitHub Free
N/A	N/A	N/A

Service & Provider	Purpose in Project	Plan / Region
N/A	N/A	N/A
N/A	N/A	N/A

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
agnes-2.5-flash (API model identifier)	Agnes API service	Pre-trained model accessed through API inference; no team training or fine-tuning.	Agnes API service terms
N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic weekly orders, product master and warehouse template	Team-created; project demo-data generator.	Synthetic	Apache-2.0 (with project)	N
Synthetic correctness, security and AI planning evaluation fixtures	Team-created; project test fixtures and evaluation scripts.	Synthetic	Apache-2.0 (with project)	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic performance dataset (10,000 order rows; 5,000 products)	Team-created; project benchmark script.	Synthetic	Apache-2.0 (with project)	N
N/A	N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A	N/A

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
OpenAI Codex	Architecture, core application code, frontend, integration, debugging, verification and documentation.	Major
DeepSeek Harness	Independent contract, API and security tests; CSV ingestion and workbook-export hardening; review.	Moderate
Replit Agent	Development, correction and testing of the Status Guard module in Replit.	Moderate
N/A	N/A	N/A
N/A	N/A	N/A

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

AI-generated output was reviewed against the project contract and independently prepared synthetic expected results. Checks cover input validation, APIs, exports, browser workflows, three-week reuse and recovery. The final repair candidate passed 178 Python tests and Windows/macOS automatic acceptance. Six bounded live-model cases passed on their first attempts; earlier failures remain recorded. Separate human Windows and Excel checks remain pending. Schema validation and explicit employee approval precede deterministic execution. The team takes responsibility for the implementation, documentation and claims.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Replit platform and Replit Agent	Used for cloud deployment of the prototype, development and testing of the Status Guard module, and part of the project's testing and data analysis.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit provides the hosting environment for the deployed prototype and supported module development, testing and data analysis. It is necessary for access through our current demonstration URL. However, the application does not fundamentally depend on Replit-specific functionality. It could be moved to another compatible hosting platform, with additional deployment configuration, storage setup and verification.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Wheat processes uploaded order files, product lists and warehouse templates. In the cloud prototype, uploaded content is sent to the backend hosted on Replit for processing. Workflow rules, approvals and run records are stored in the deployed application's database. When AI planning is enabled, bounded file profiles and samples are also sent to the Agnes API service. These samples may contain uploaded information, so the demonstration uses only team-created synthetic data without personal information. Users should not upload confidential or personal data to the demonstration. File persistence, deletion behaviour, access controls and the retention policies of Replit and the AI provider must be verified before real business use. We do not yet claim a fixed automatic deletion period.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The AI may misunderstand column names, product descriptions, order statuses or quantity units. It may also suggest incorrect or incomplete processing rules. To reduce these risks, the application checks proposed workflows through Pydantic and application validation, then requires users to review and approve the rules before execution. Only bundled processing operations are allowed, and the system does not execute arbitrary AI-generated code. Exceptions and row-level processing records help users check and correct results. AI planning also depends on the external provider's availability, usage limits and model behaviour. Without API credentials, deterministic workflow setup remains available. Live-model reliability and Windows verification remain outstanding. The prototype is deployed on Replit, but concurrent multi-user use and isolation between users require further verification. Orbyte handoff is simulated. Further testing is needed before wider deployment or use with real business data.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature] 

[Date]2026/10/24